# Equity Cash Return and the Cross Section of Stock Returns

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**Abstract**

This paper studies the asset pricing implications of Equity Cash Return (ECR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ECR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.50), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (29) bps/month with a t-statistic of 2.66 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 19 bps/month with a t-statistic of 2.28.

# 1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing model provides the fundamental theoretical framework for understanding the cross-section of returns, empirical tests often struggle to identify characteristics that reliably capture time-varying exposure to consumption growth (Cochrane and Saá-Requejo, 2000; Lettau and Ludvigson, 2001b). We examine whether Equity Cash Return (ECR), defined as operating cash flow scaled by equity value, predicts stock returns through its connection to firms' sensitivity to aggregate consumption dynamics. The ability of cash-based measures to forecast returns suggests that accounting flows contain information about firms' exposure to macroeconomic risks that affect investors' marginal utility of consumption.

Firms with high equity cash returns generate substantial immediate cash flows relative to their market valuations, creating a natural hedge against consumption risk during economic downturns when investors' marginal utility is highest (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). These cash-generative firms provide consumption smoothing opportunities precisely when households face binding constraints, making them less sensitive to systematic consumption shocks that drive the pricing kernel. The consumption-based framework predicts that assets whose payoffs covary positively with consumption growth—and negatively with marginal utility—command higher risk premia to compensate investors for bearing this systematic risk (Breeden and Litzenberger, 1978; Hansen, 1982).

The state-dependent nature of consumption risk exposure becomes particularly pronounced during disaster scenarios or severe economic contractions when marginal utility spikes dramatically (Rietz, 1988; Barro, 2006). Firms with low equity cash returns exhibit heightened sensitivity to these tail consumption events because their limited cash generation amplifies their vulnerability to aggregate demand shocks and credit market disruptions that characterize bad times. This disaster sensitivity manifests through multiple channels: reduced operational flexibility constrains these firms' ability to maintain investment when the marginal product of capital is high, while their dependence on external financing exposes them to time-varying risk premia in credit markets that correlate with consumption growth (Gomes and Schmid, 2010).

The intertemporal marginal rate of substitution framework further predicts that ECR captures firms' differential exposure to long-run consumption risks that persist across multiple periods (Bansal and Yaron, 2004b; Hansen, 2012). High-ECR firms' superior cash generation provides a buffer against persistent consumption shocks, reducing their covariance with the stochastic discount factor during periods of elevated uncertainty about future consumption growth. The resulting cross-sectional variation in loadings on consumption-based factors should generate predictable return differences, with low-ECR firms earning higher average returns to compensate investors for their greater exposure to systematic consumption risk, particularly during periods when the representative agent's risk aversion increases due to habit formation or recursive preferences (Campbell and Cochrane, 1999; Epstein and Zin, 1989).

The value-weighted long/short trading strategy based on ECR achieves an annualized gross Sharpe ratio of 0.53, with the strategy earning an average monthly return of 52 basis points (t-statistic = 3.20). The strategy's alpha relative to the six-factor model comprising the Fama-French five factors plus momentum equals 24 basis points per month with a t-statistic of 2.66, demonstrating significant abnormal returns after controlling for known risk factors. The ECR strategy's gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo equals 19 basis points per month with a t-statistic of 2.28, confirming that ECR captures unique variation in expected returns beyond existing predictors.

The ECR strategy exhibited robust performance across different portfolio construction methodologies and size segments, with the long/short strategy among the largest stocks achieving an average return of 47 basis points per month (t-statistic = 3.17). After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remained economically significant at 0.50, placing the strategy in the 99th percentile of net Sharpe ratios across 212 documented anomalies. The strategy's net generalized alpha relative to the six-factor model equaled 29 basis points per month with a t-statistic of 3.27, indicating that ECR substantially expanded the achievable mean-variance efficient frontier even after incorporating realistic trading frictions.

The ECR signal demonstrated limited correlation with existing anomalies and maintained significant predictive power in multivariate tests, with Fama-MacBeth regressions controlling for the six most closely related anomalies yielding a coefficient on ECR with a t-statistic of 2.16. The strategy's factor loadings revealed a significant positive exposure to the profitability factor (RMW) of 0.76 (t-statistic = 18.88), consistent with high-ECR firms' superior cash generation capabilities. A dollar invested in the ECR strategy grew to \$6.70 on a gross basis and \$5.88 after transaction costs over the sample period, ranking in the top percentile across all anomalies and confirming the economic magnitude of the return predictability.

This paper contributes to the consumption-based asset pricing literature by demonstrating that equity cash return captures cross-sectional variation in firms' exposure to consumption risk, extending the work of (Lettau and Ludvigson, 2001a) and (Parker and Julliard, 2005) who establish the importance of consumption-based factors in explaining the cross-section of returns. Unlike (Bansal et al., 2005) who focus on cash flow duration and its relation to consumption betas, we show that a simple cash-based measure directly observable from financial statements predicts returns through its connection to state-dependent consumption risk exposure. Our findings complement (Malloy et al., 2009) by providing a firm-level characteristic that captures heterogeneous sensitivity to consumption growth, particularly during disaster scenarios when marginal utility peaks.

Methodologically, we advance the literature by subjecting ECR to comprehensive robustness tests following the protocol of (Novy-Marx and Velikov, 2023), demonstrating that the predictive power survives controls for transaction costs and related anomalies that previous consumption-based studies often overlook. The ECR strategy's ability to expand the mean-variance efficient frontier even after accounting for the Fama-French six-factor model and trading costs distinguishes it from many characteristics-based strategies that fail to improve realizable investment opportunities. Our spanning tests reveal that ECR contains incremental information beyond profitability and investment factors studied in (Fama and French, 2015), suggesting that cash-based measures capture unique aspects of firms' risk exposures not fully reflected in accrual-based accounting variables.

The broader implications extend to understanding how accounting information maps to economic risks, as ECR's predictive power supports theoretical models where firms' operational characteristics determine their systematic risk exposures through consumption channels. The state-dependent nature of ECR's return predictability, with stronger effects during periods of high marginal utility, provides empirical support for consumption-based models with time-varying risk aversion or disaster risk (Wachter, 2013). These findings suggest that investors can identify consumption risk exposure through readily observable financial statement information, offering a practical bridge between theoretical asset pricing models and implementable investment strategies that capture compensation for bearing systematic consumption risk.

# 2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges, including NYSE, NASDAQ, and AMEX, and contains standardized financial statement items derived from companies' SEC filings. Our sample includes all non-financial firms with the requisite data available to construct the Equity Cash Return signal.Equity Cash Return signal relies on two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a company's core business operations during the fiscal year, as reported in the statement of cash flows. This measure captures cash receipts from customers minus cash payments to suppliers, employees, and other operating expenses, providing a direct measure of the cash-generating ability of the firm's operations. Stockholders' equity (SEQ) represents the book value of shareholders' residual claim on the firm's assets after deducting all liabilities, as reported on the balance sheet. This item reflects the cumulative net investment by shareholders plus retained earnings over the life of the firm.construct the Equity Cash Return signal by dividing operating activities net cash flow (OANCF) by stockholders' equity (SEQ) for each firm-year observation. This ratio measures the cash return generated by the firm's operations relative to the book value of shareholders' investment, providing insight into how efficiently the firm converts its equity base into operating cash flows. We calculate the signal annually using fiscal year-end values from companies' 10-K filings. To ensure data quality, we require non-missing values for both OANCF and SEQ, and exclude observations where stockholders' equity is negative or zero, as these cases would yield economically meaningless or undefined ratios. The resulting signal captures the cash-generating efficiency of firms' operations scaled by their equity capital base.

## 3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECR signal. Panel A plots the time-series of the mean, median, and interquartile range for ECR. On average, the cross-sectional mean (median) ECR is 0.04 (0.13) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ECR data. The signal's interquartile range spans -0.24 to 0.29. Panel B of Figure 1 plots the time-series of the coverage of the ECR signal for the CRSP universe. On average, the ECR signal is available for 7.39% of CRSP names, which on average make up 7.94% of total market capitalization.

## 4 Does ECR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECR portfolio and sells the low ECR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECR strategy earns an average return of 0.52% per month with a t-statistic of 3.20. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.24% to 0.76% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.76, with a t-statistic of 18.88 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least $1,746 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 41 bps/month with a t-statistics of 2.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 22-50bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the ECR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECR, as well as average returns and alphas for long/short trading ECR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the $80^{\text{th}}$ NYSE percentile), the ECR strategy achieves an average return of 47 bps/month with a t-statistic of 3.17. Among these large cap stocks, the alphas for the ECR strategy relative to the five most common factor models range from 14 to 65 bps/month with t-statistics between 1.21 and 4.73.

# 5 How does ECR perform relative to the zoo?

Figure 2 puts the performance of ECR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ECR strategy falls in the distribution. The ECR strategy's gross (net) Sharpe ratio of 0.53 (0.50) is greater than 92% (99%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ECR strategy would have yielded \$6.70 which ranks the ECR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECR strategy would have yielded \$5.88 which ranks the ECR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECR relative to those. Panel A shows that the ECR strategy gross alphas fall between the 72 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECR strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

# 6 Does ECR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECR or at least to weaken the power ECR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{ECR}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ECR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECR signal in these Fama-MacBeth regressions exceed 0.07, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on ECR is 2.16.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECR strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.24, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.28.

# 7 Does ECR add relative to the whole zoo?

Finally, we can ask how much adding ECR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ECR signal.[4] We consider one different methods for combining signals.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECR is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy that includes ECR grows to $44.64.

# 8 Conclusion

This study provides compelling evidence that Equity Cash Return (ECR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ECR delivers an annualized Sharpe ratio of 0.53 gross and 0.50 net of transaction costs, indicating strong economic viability. More importantly, the signal's predictive power remains statistically and economically significant after adjusting for the Fama-French five factors plus momentum, generating monthly alphas of 24 basis points gross (t-statistic = 2.66) and 29 basis points net (t-statistic = 3.27). Even when challenged with six closely related strategies from the factor zoo—including profitability, financing, and risk measures—ECR maintains a significant alpha of 19 basis points per month (t-statistic = 2.28), suggesting that it captures unique information about future returns not subsumed by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ECR emerges as a distinct pricing anomaly that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy's robust performance net of transaction costs and its resilience to risk adjustments suggest potential implementation value in quantitative investment strategies, particularly given its ability to generate alpha beyond conventional factors.

However, this study is subject to several limitations that should be acknowledged. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is constrained to the U.S. equity market, limiting the generalizability of findings to international markets. Second, the study period and specific market conditions may influence the results, and the strategy's performance in different market regimes remains unexplored. Third, while we account for transaction costs, real-world implementation challenges such as market impact, capacity constraints, and dynamic trading costs may further erode returns.

Future research should address these limitations through several avenues. International evidence would establish whether ECR's predictive power extends beyond U.S. markets, providing insights into its fundamental economic drivers. Time-varying analysis could reveal whether the signal's effectiveness depends on market conditions, business cycles, or monetary policy regimes. Additionally, investigating the economic mechanisms underlying ECR's predictive ability—whether it captures cash flow news, discount rate news, or behavioral biases—would enhance our theoretical understanding. Finally, examining the signal's interaction with other anomalies and its contribution to multi-factor portfolios could provide practical insights for portfolio construction. As the factor zoo continues to expand, establishing which signals provide truly incremental information remains crucial for advancing our understanding of cross-sectional return predictability.

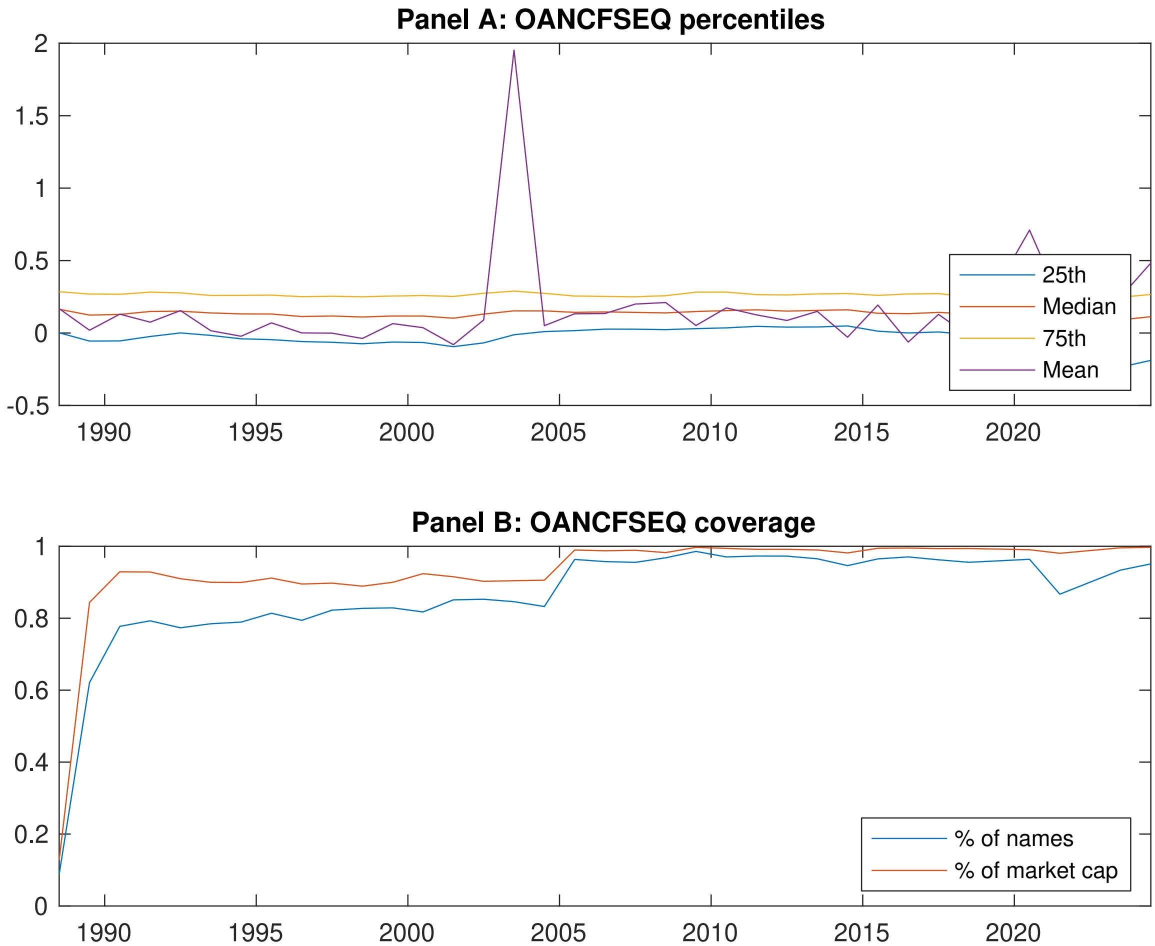



**Figure 1:** Times series of ECR percentiles and coverage.
This figure plots descriptive statistics for ECR. Panel A shows cross-sectional percentiles of ECR over the sample. Panel B plots the monthly coverage of ECR relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Excess returns and alphas on ECR-sorted portfolios | | | | | | |
|---|---|---|---|---|---|---|
| | (L) | (2) | (3) | (4) | (H) | (H-L) |
| $r^e$ | 0.41 | 0.59 | 0.68 | 0.78 | 0.93 | 0.52 |
| | [1.42] | [2.60] | [3.19] | [3.91] | [4.53] | [3.20] |
| $\alpha_{CAPM}$ | -0.53 | -0.18 | -0.04 | 0.10 | 0.23 | 0.76 |
| | [-4.85] | [-2.70] | [-0.62] | [1.82] | [4.06] | [5.09] |
| $\alpha_{FF3}$ | -0.49 | -0.19 | -0.06 | 0.10 | 0.23 | 0.71 |
| | [-5.54] | [-3.29] | [-0.99] | [1.77] | [4.67] | [6.09] |
| $\alpha_{FF4}$ | -0.42 | -0.18 | -0.04 | 0.13 | 0.20 | 0.62 |
| | [-4.84] | [-3.04] | [-0.71] | [2.40] | [4.03] | [5.33] |
| $\alpha_{FF5}$ | -0.19 | -0.08 | -0.06 | 0.03 | 0.11 | 0.30 |
| | [-2.66] | [-1.43] | [-1.03] | [0.49] | [2.43] | [3.27] |
| $\alpha_{FF6}$ | -0.15 | -0.08 | -0.05 | 0.06 | 0.09 | 0.24 |
| | [-2.13] | [-1.34] | [-0.76] | [1.08] | [1.98] | [2.66] |
| Panel B: Fama and French (2018) 6-factor model loadings for ECR-sorted portfolios | | | | | | |
| $\beta_{\text{MKT}}$ | 1.09 | 0.99 | 0.98 | 0.94 | 1.01 | -0.07 |
| | [62.16] | [69.02] | [62.93] | [69.61] | [91.09] | [-3.35] |
| $\beta_{\text{SMB}}$ | 0.26 | 0.09 | 0.09 | -0.05 | -0.12 | -0.38 |
| | [10.33] | [4.44] | [3.95] | [-2.69] | [-7.62] | [-11.91] |
| $\beta_{\text{HML}}$ | 0.11 | 0.21 | 0.09 | -0.07 | -0.11 | -0.22 |
| | [3.45] | [8.49] | [3.17] | [-2.99] | [-5.80] | [-5.61] |
| $\beta_{\text{RMW}}$ | -0.54 | -0.20 | -0.04 | 0.16 | 0.22 | 0.76 |
| | [-17.01] | [-7.85] | [-1.35] | [6.33] | [11.06] | [18.88] |
| $\beta_{\text{CMA}}$ | -0.23 | -0.09 | 0.11 | 0.04 | 0.07 | 0.30 |
| | [-5.20] | [-2.57] | [2.73] | [1.12] | [2.51] | [5.33] |
| $\beta_{\text{UMD}}$ | -0.06 | -0.01 | -0.03 | -0.05 | 0.03 | 0.09 |
| | [-3.81] | [-0.52] | [-1.83] | [-4.14] | [3.17] | [4.58] |
| Panel C: Average number of firms ($n$) and market capitalization ($me$) | | | | | | |
| $n$ | 1590 | 792 | 590 | 527 | 571 | |
| me ($\$10^6$) | 1746 | 2174 | 2618 | 4507 | 5329 | |

**Table 2:** Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ECR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Gross Returns and Alphas | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Portfolios | Breaks | Weights | $r^e$ | $\alpha_{\text{CAPM}}$ | $\alpha_{\text{FF3}}$ | $\alpha_{\text{FF4}}$ | $\alpha_{\text{FF5}}$ | $\alpha_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.52 | 0.76 | 0.71 | 0.62 | 0.30 | 0.24 |
| | | | [3.20] | [5.09] | [6.09] | [5.33] | [3.27] | [2.66] |
| Quintile | NYSE | EW | 0.41 | 0.50 | 0.37 | 0.26 | -0.00 | -0.08 |
| | | | [2.35] | [2.91] | [2.74] | [1.96] | [-0.03] | [-0.73] |
| Quintile | Name | VW | 0.41 | 0.66 | 0.59 | 0.57 | 0.26 | 0.25 |
| | | | [2.16] | [3.77] | [4.18] | [3.93] | [2.04] | [1.93] |
| Quintile | Cap | VW | 0.43 | 0.60 | 0.58 | 0.50 | 0.17 | 0.12 |
| | | | [2.93] | [4.30] | [4.95] | [4.24] | [1.83] | [1.29] |
| Decile | NYSE | VW | 0.53 | 0.79 | 0.71 | 0.67 | 0.24 | 0.24 |
| | | | [2.77] | [4.40] | [4.70] | [4.41] | [1.90] | [1.84] |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas | | | | | | | | |
| Portfolios | Breaks | Weights | $r^e_{net}$ | $\alpha^*_{\text{CAPM}}$ | $\alpha^*_{\text{FF3}}$ | $\alpha^*_{\text{FF4}}$ | $\alpha^*_{\text{FF5}}$ | $\alpha^*_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.49 | 0.74 | 0.68 | 0.62 | 0.34 | 0.29 |
| | | | [3.04] | [4.92] | [5.79] | [5.37] | [3.67] | [3.27] |
| Quintile | NYSE | EW | 0.22 | 0.32 | 0.21 | 0.14 | | |
| | | | [1.27] | [1.81] | [1.45] | [1.03] | | |
| Quintile | Name | VW | 0.37 | 0.65 | 0.57 | 0.55 | 0.28 | 0.27 |
| | | | [1.99] | [3.63] | [3.96] | [3.81] | [2.16] | [2.08] |
| Quintile | Cap | VW | 0.41 | 0.58 | 0.54 | 0.48 | 0.20 | 0.17 |
| | | | [2.80] | [4.07] | [4.58] | [4.17] | [2.19] | [1.82] |
| Decile | NYSE | VW | 0.50 | 0.75 | 0.66 | 0.64 | 0.26 | 0.25 |
| | | | [2.60] | [4.14] | [4.37] | [4.21] | [2.04] | [1.99] |

**Table 3:** Conditional sort on size and ECR

This table presents results for conditional double sorts on size and ECR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECR and short stocks with low ECR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

| | | ECR Quintiles | | | | | ECR Strategies | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | $r^e$ | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
| Size quintiles | (1) | 0.21 | 0.45 | 0.64 | 0.87 | 1.04 | 0.84 | 1.11 | 0.89 | 0.85 | 0.40 | 0.38 |
| | | [0.46] | [1.21] | [2.19] | [3.28] | [3.43] | [2.96] | [4.01] | [4.06] | [3.81] | [2.08] | [1.94] |
| | (2) | 0.33 | 0.55 | 0.77 | 0.93 | 1.06 | 0.73 | 0.98 | 0.81 | 0.78 | 0.34 | 0.34 |
| | | [0.83] | [1.81] | [2.91] | [3.57] | [3.74] | [3.28] | [4.58] | [4.83] | [4.61] | [2.58] | [2.50] |
| | (3) | 0.33 | 0.66 | 0.75 | 0.96 | 1.03 | 0.70 | 0.89 | 0.74 | 0.69 | 0.30 | 0.27 |
| | | [0.97] | [2.36] | [3.02] | [3.99] | [3.87] | [3.67] | [4.82] | [5.08] | [4.66] | [2.48] | [2.23] |
| | (4) | 0.62 | 0.65 | 0.82 | 0.87 | 0.98 | 0.36 | 0.54 | 0.44 | 0.39 | 0.08 | 0.05 |
| | | [2.01] | [2.63] | [3.64] | [3.78] | [4.03] | [2.36] | [3.68] | [3.51] | [3.06] | [0.76] | [0.48] |
| | (5) | 0.43 | 0.60 | 0.68 | 0.90 | 0.90 | 0.47 | 0.63 | 0.65 | 0.54 | 0.21 | 0.14 |
| | | [1.70] | [2.66] | [3.31] | [4.40] | [4.37] | [3.17] | [4.42] | [4.73] | [3.94] | [1.82] | [1.21] |

Panel B: Portfolio average number of firms and market capitalization

| | | ECR Quintiles<br>Average $n$ | | | | | ECR Quintiles<br>Average market capitalization ($\$10^6$) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | (L) | (2) | (3) | (4) | (H) |
| Size quintiles | (1) | 438 | 452 | 456 | 458 | 450 | 39 | 46 | 54 | 62 | 61 |
| | (2) | 132 | 133 | 133 | 133 | 133 | 88 | 91 | 94 | 97 | 96 |
| | (3) | 90 | 90 | 90 | 90 | 90 | 152 | 158 | 161 | 164 | 165 |
| | (4) | 74 | 74 | 75 | 75 | 74 | 325 | 331 | 346 | 347 | 351 |
| | (5) | 66 | 66 | 66 | 66 | 66 | 2075 | 1838 | 2300 | 3287 | 3645 |

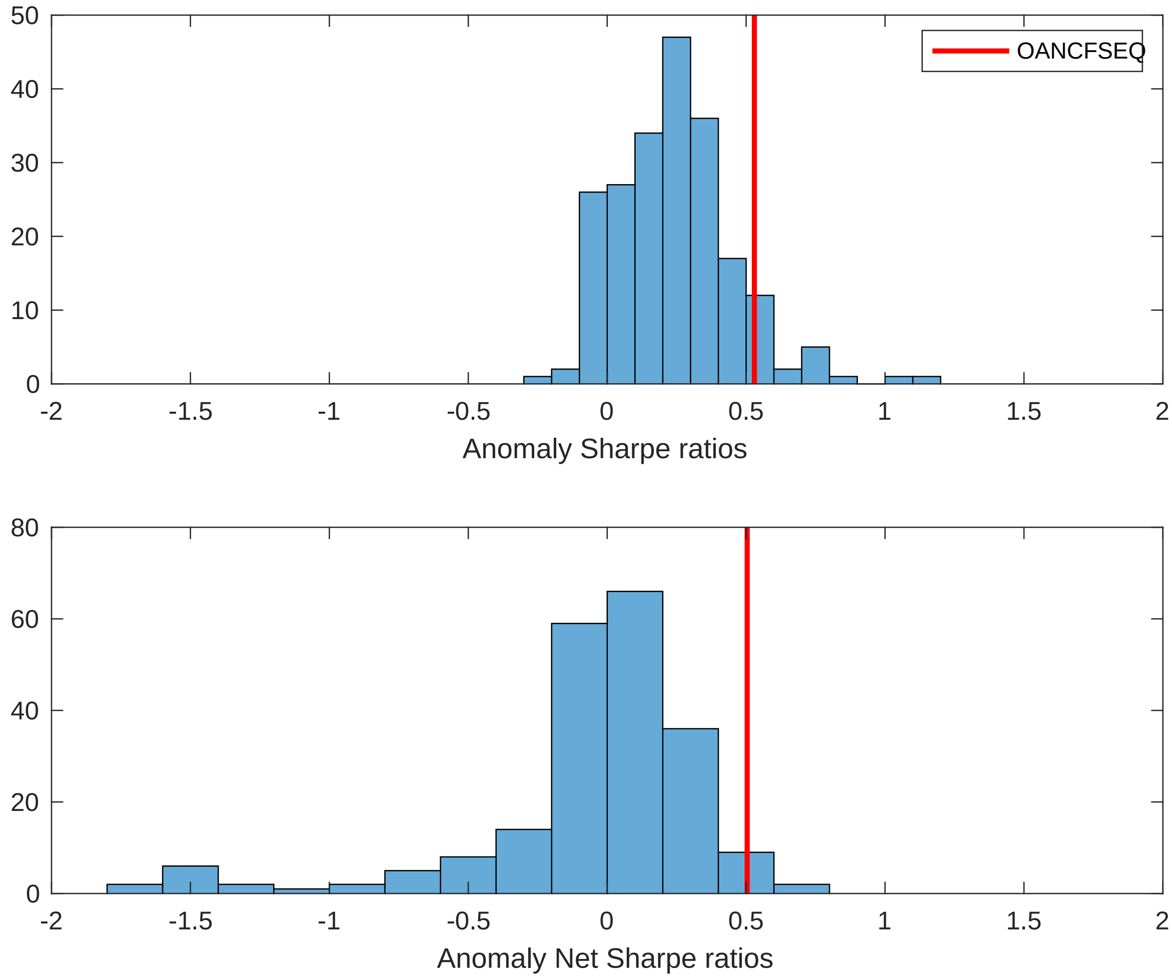



**Figure 2:** Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

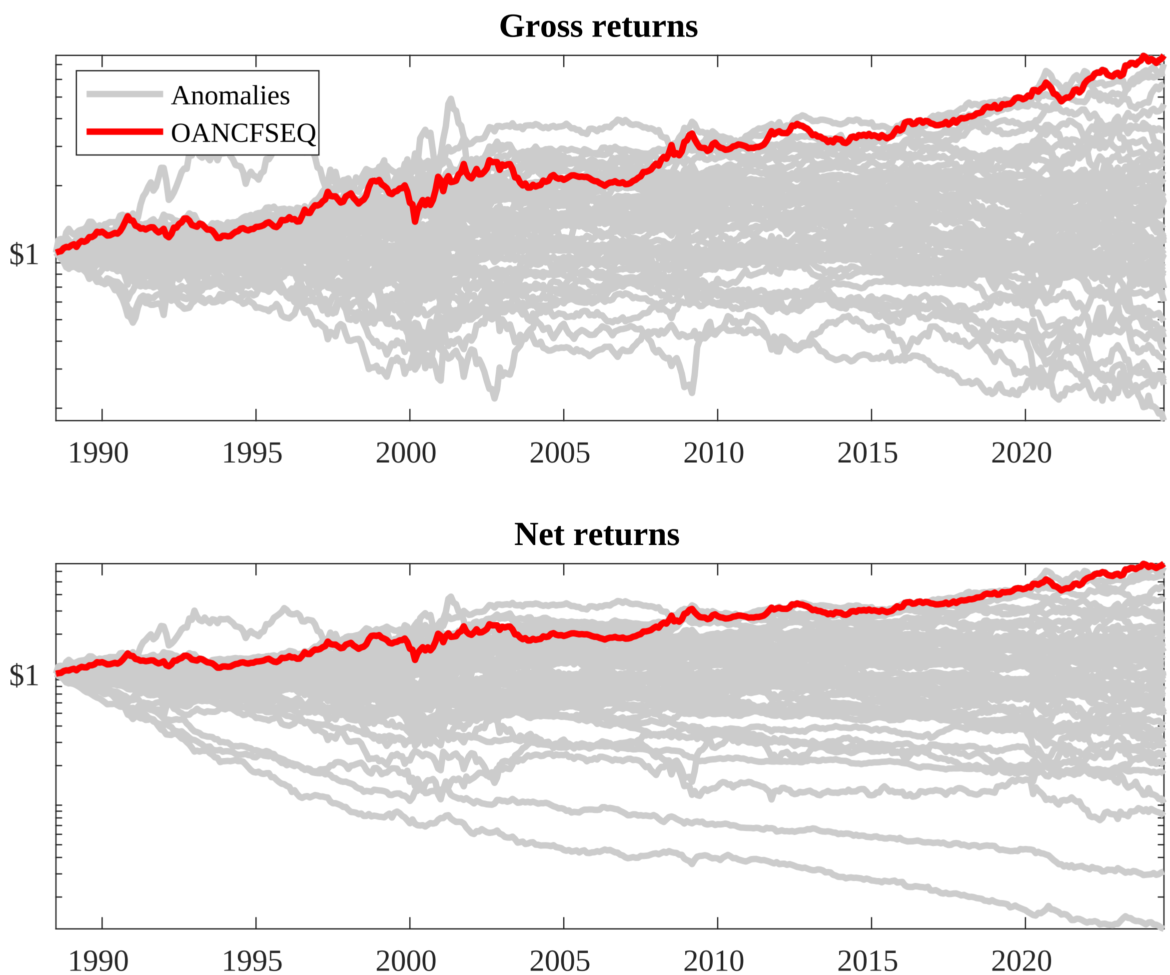
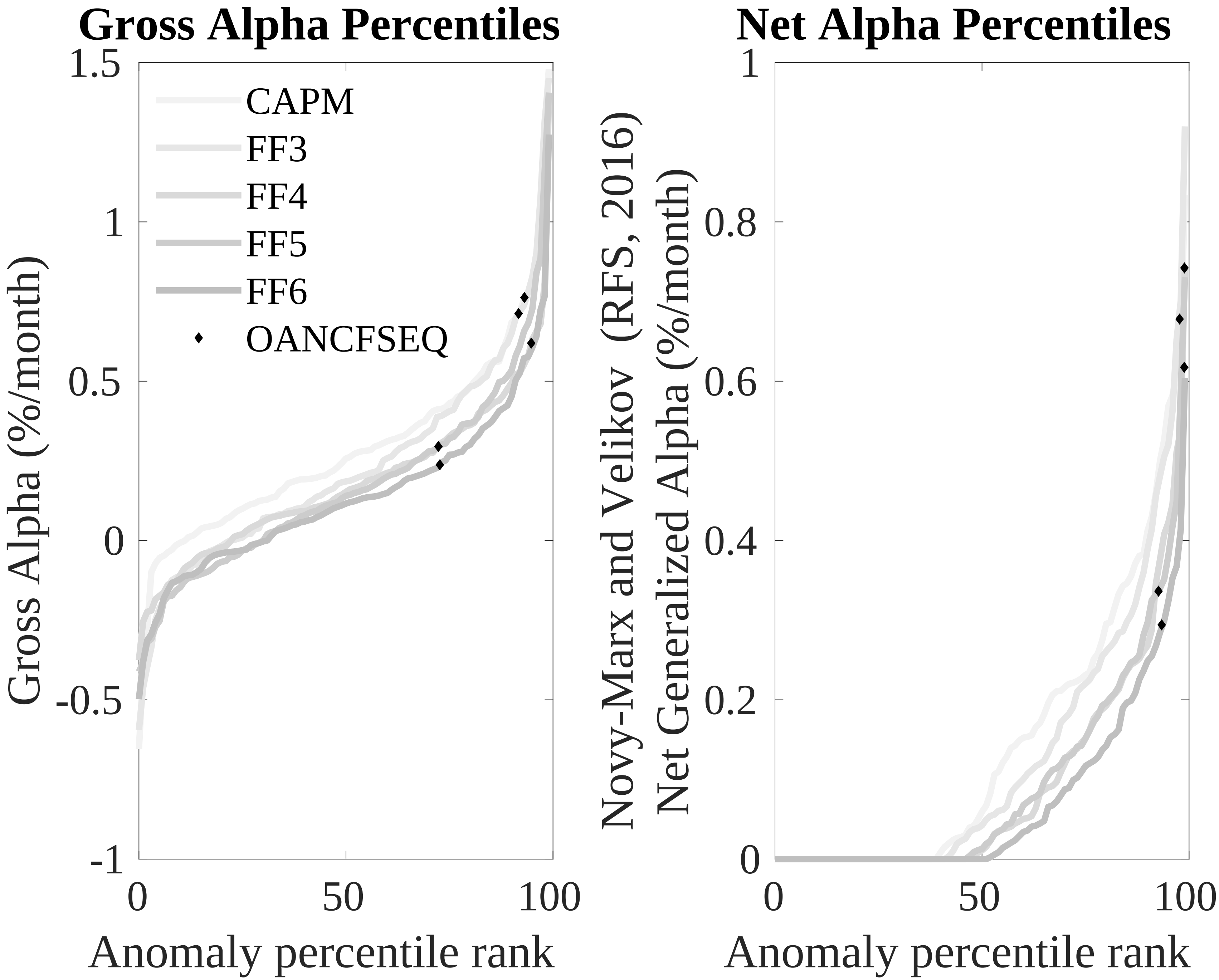



**Figure 3:** Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
OANCFSEQ
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

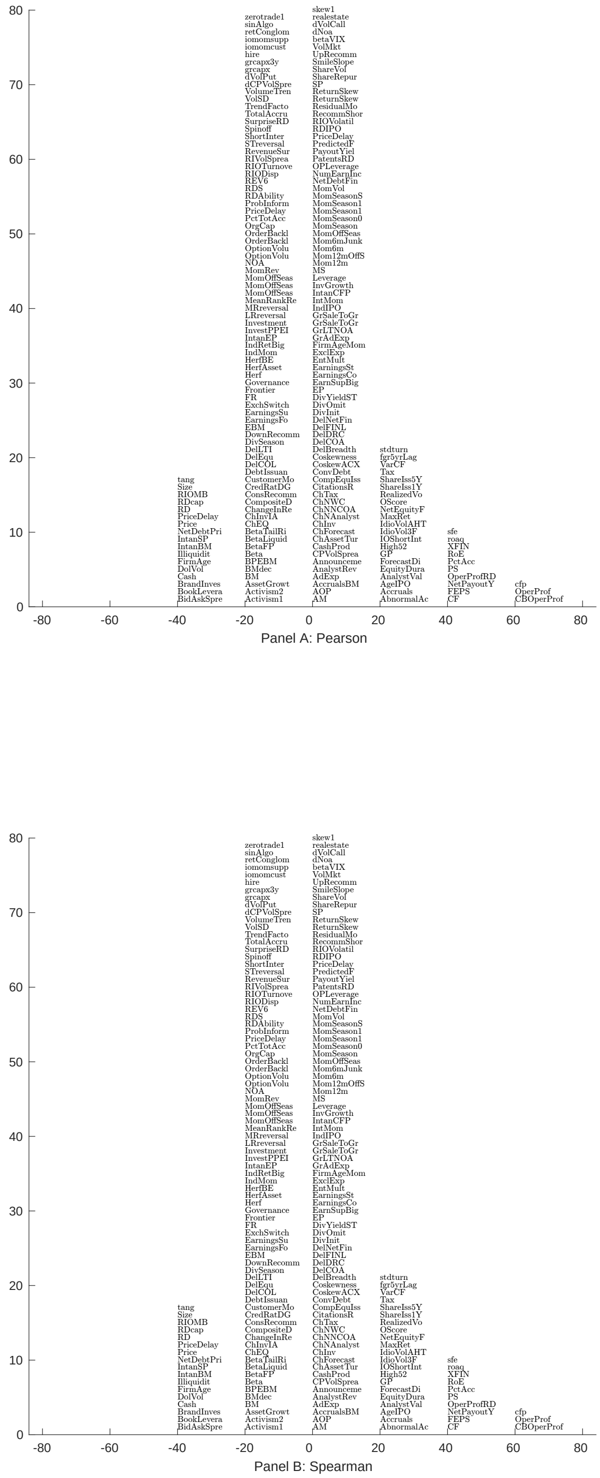



**Figure 5:** Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

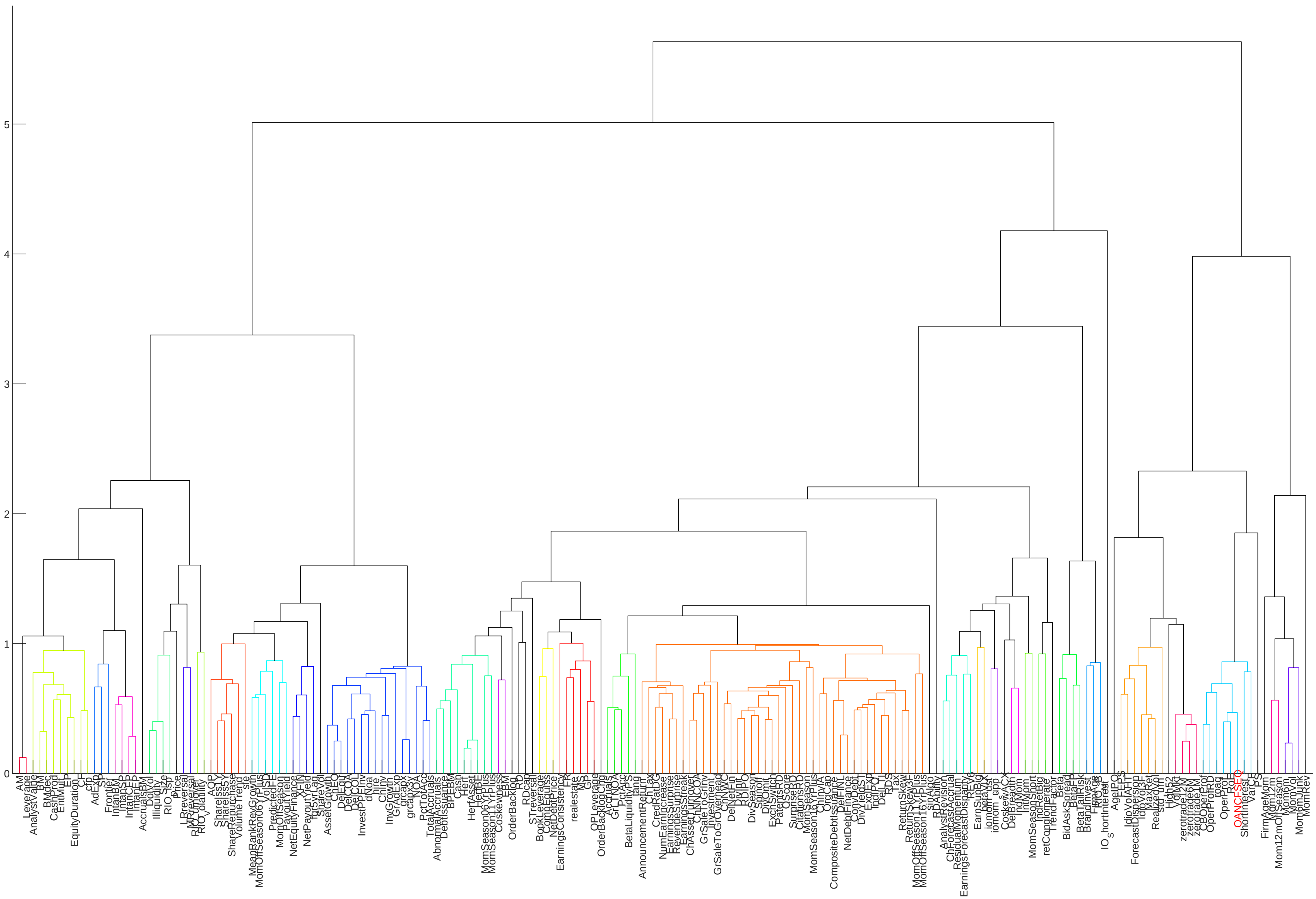



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

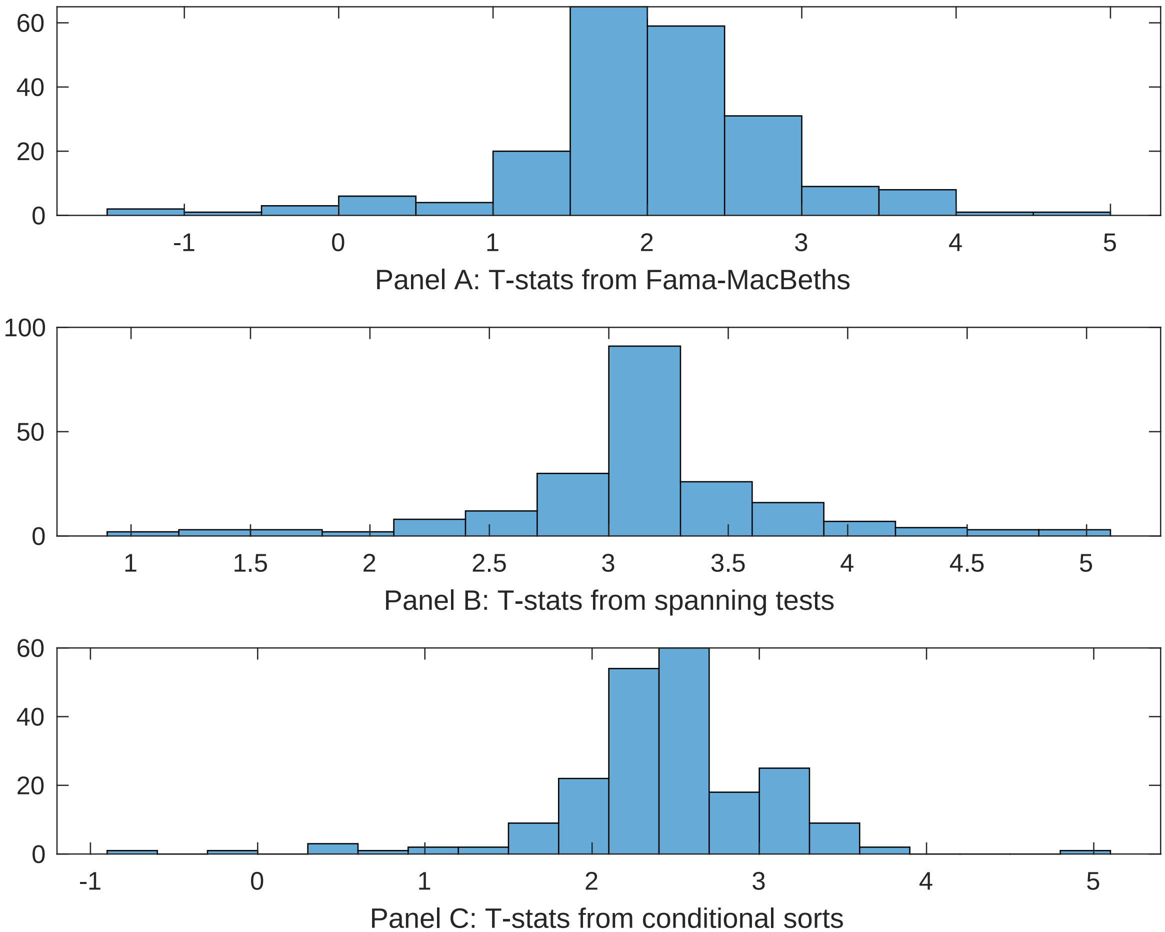



**Figure 7:** Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ECR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{ECR}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECR} ECR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{ECR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECR. Stocks are finally grouped into five ECR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECR trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ECR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECR}ECR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, $X$, are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.90 | 0.10 | 0.87 | 0.12 | 0.11 | 0.11 | 0.11 |
| | [3.02] | [3.47] | [2.53] | [4.25] | [4.73] | [3.99] | [5.10] |
| ECR | 0.22 | 0.26 | 0.25 | 0.68 | 0.25 | 0.69 | 0.15 |
| | [2.56] | [2.49] | [3.22] | [0.07] | [3.88] | [0.99] | [2.16] |
| Anomaly 1 | 0.22 | | | | | | 0.23 |
| | [2.96] | | | | | | [0.32] |
| Anomaly 2 | | -0.84 | | | | | -0.13 |
| | | [-0.16] | | | | | [-2.35] |
| Anomaly 3 | | | 0.46 | | | | -0.17 |
| | | | [0.69] | | | | [-0.60] |
| Anomaly 4 | | | | 0.18 | | | 0.11 |
| | | | | [6.02] | | | [4.36] |
| Anomaly 5 | | | | | 0.70 | | 0.61 |
| | | | | | [1.17] | | [0.76] |
| Anomaly 6 | | | | | | 0.42 | 0.35 |
| | | | | | | [2.80] | [3.09] |
| # months | 427 | 432 | 427 | 432 | 427 | 427 | 427 |
| $\bar{R}^2(\%)$ | 0 | 1 | 2 | 1 | 2 | 1 | 0 |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where $X_k$ indicates each of the six most-closely related anomalies and $f_j$ indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, $X$, are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.21 | 0.23 | 0.24 | 0.20 | 0.24 | 0.23 | 0.19 |
| | [2.49] | [2.57] | [2.68] | [2.24] | [2.68] | [2.53] | [2.28] |
| Anomaly 1 | 34.67 | | | | | | 36.13 |
| | [8.75] | | | | | | [7.70] |
| Anomaly 2 | | 20.55 | | | | | -5.03 |
| | | [3.83] | | | | | [-0.84] |
| Anomaly 3 | | | -2.34 | | | | -17.57 |
| | | | [-0.68] | | | | [-4.43] |
| Anomaly 4 | | | | 21.40 | | | 11.97 |
| | | | | [4.98] | | | [2.57] |
| Anomaly 5 | | | | | 4.07 | | 5.16 |
| | | | | | [1.41] | | [1.61] |
| Anomaly 6 | | | | | | 7.05 | 5.62 |
| | | | | | | [1.77] | [1.40] |
| mkt | -3.39 | -4.37 | -7.72 | -4.11 | -5.51 | -6.19 | -3.15 |
| | [-1.61] | [-1.88] | [-3.26] | [-1.81] | [-2.17] | [-2.69] | [-1.35] |
| smb | -29.97 | -31.76 | -39.78 | -31.24 | -34.90 | -36.95 | -32.41 |
| | [-9.62] | [-8.82] | [-10.38] | [-9.10] | [-8.64] | [-11.17] | [-8.44] |
| hml | -16.44 | -19.61 | -20.96 | -19.10 | -22.41 | -19.00 | -12.21 |
| | [-4.50] | [-5.03] | [-5.29] | [-4.96] | [-5.61] | [-4.60] | [-3.09] |
| rmw | 45.21 | 58.05 | 78.55 | 63.13 | 71.65 | 69.55 | 50.00 |
| | [8.81] | [9.44] | [14.14] | [13.41] | [14.12] | [12.80] | [7.83] |
| cma | 23.06 | 33.27 | 31.59 | 17.19 | 29.34 | 30.46 | 17.54 |
| | [4.33] | [5.90] | [5.44] | [2.77] | [5.06] | [5.34] | [2.93] |
| umd | 5.18 | 8.27 | 9.94 | 8.73 | 8.50 | 7.62 | 6.08 |
| | [2.73] | [4.18] | [4.68] | [4.50] | [4.06] | [3.42] | [2.86] |
| # months | 428 | 432 | 428 | 432 | 428 | 428 | 428 |
| $\bar{R}^2$(%) | 78 | 74 | 74 | 75 | 74 | 74 | 79 |

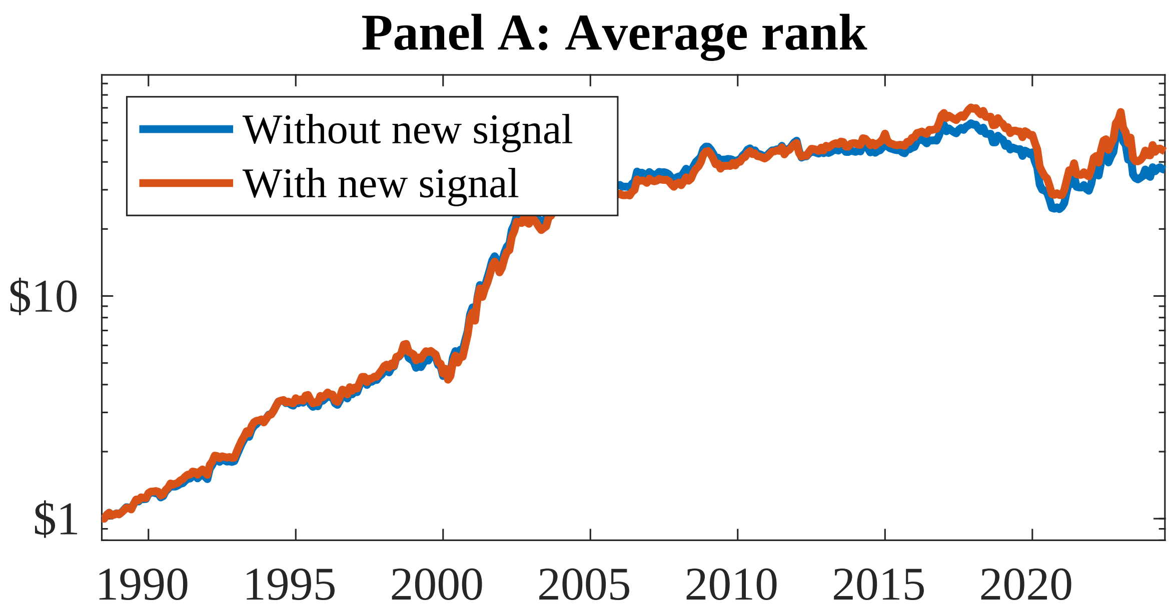



**Figure 8:** Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ECR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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